1) a)

Prove true for $n = 1$:

RHS: $\cos 1\theta + \mathrm{i}\sin 1\theta$ which is the LHS so true for $n = 1$

Assume true for $n = k$:

$(\cos\theta + \mathrm{i}\sin\theta)^k = \cos k\theta + \mathrm{i}\sin k\theta$

Prove true for $n = k + 1$:

$(\cos\theta + \mathrm{i}\sin\theta)^{k+1} = (\cos\theta + \mathrm{i}\sin\theta)(\cos\theta + \mathrm{i}\sin\theta)^k$

$= (\cos\theta + \mathrm{i}\sin\theta)(\cos k\theta + \mathrm{i}\sin k\theta)$

$= \cos\theta\cos k\theta - \sin\theta\sin k\theta + \mathrm{i}\sin\theta\cos k\theta + \mathrm{i}\cos\theta\sin k\theta$

$= (\cos(\theta + k\theta)) + \mathrm{i}(\sin(\theta + k\theta))$

$= (\cos(k+1)\theta) + \mathrm{i}(\sin(k+1)\theta)$

Which is $(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

1) b)

$\frac{1}{\cos\theta+\mathrm{i}\sin\theta} \times \frac{\cos\theta-\mathrm{i}\sin\theta}{\cos\theta-\mathrm{i}\sin\theta} = \frac{\cos\theta-\mathrm{i}\sin\theta}{\cos^2\theta+\sin^2\theta} = \cos\theta - \mathrm{i}\sin\theta$

Hence

$\frac{1}{(\cos\theta+\mathrm{i}\sin\theta)^n} = (\cos\theta - \mathrm{i}\sin\theta)^n$

But since $(\cos x + \mathrm{i}\sin x)^n = \cos nx + \mathrm{i}\sin nx$ then set $x = -\theta$ so that

$(\cos(-\theta) + \mathrm{i}\sin(-\theta))^n = \cos n(-\theta) + \mathrm{i}\sin n(-\theta)$

$\Rightarrow (\cos\theta - \mathrm{i}\sin\theta)^n = \cos n\theta - \mathrm{i}\sin n\theta$

So that $(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$ for all integers $n < 0$

When $n = 0$:

$(\cos\theta + \mathrm{i}\sin\theta)^0 = 1$ and $\cos 0 + \mathrm{i}\sin 0 = 1$

$\therefore (\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$ for all integers n

2)

Prove true for $n = -1$:

Note that: $\frac{1}{\cos\theta+\mathrm{i}\sin\theta} \times \frac{\cos\theta-\mathrm{i}\sin\theta}{\cos\theta-\mathrm{i}\sin\theta} = \frac{\cos\theta-\mathrm{i}\sin\theta}{\cos^2\theta+\sin^2\theta} = \cos\theta - \mathrm{i}\sin\theta$

RHS: $\cos(-\theta) + \mathrm{i}\sin(-\theta) = \cos\theta - \mathrm{i}\sin\theta = \frac{1}{\cos\theta+\mathrm{i}\sin\theta}$ which is the LHS so true for $n = -1$

Assume true for $n = k$:

$(\cos\theta + \mathrm{i}\sin\theta)^k = \cos k\theta + \mathrm{i}\sin k\theta$

Prove true for $n = k - 1$:

$(\cos\theta + \mathrm{i}\sin\theta)^{k-1} = (\cos\theta + \mathrm{i}\sin\theta)^k(\cos\theta - \mathrm{i}\sin\theta)$

$= (\cos k\theta + \mathrm{i}\sin k\theta)(\cos\theta - \mathrm{i}\sin\theta)$

$= \cos k\theta\cos\theta + \sin k\theta\sin\theta + \mathrm{i}(\sin k\theta\cos\theta - \cos k\theta\sin\theta)$

$= (\cos(k\theta - \theta)) + \mathrm{i}(\sin(k\theta - \theta))$

$= (\cos(k-1)\theta) + \mathrm{i}(\sin(k-1)\theta)$

Which is $(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$ with $k - 1$ substituted in.

As the statement is true for $n = -1$, and if the statement is true for $n = k$ then the statement is true for $n = k - 1$, so the statement is true for all integers $n \leq -1$

3)

Prove true for $n = 1$:

LHS: $\frac{1}{3}$

RHS: $\frac{1}{2\times1+1} = \frac{1}{3}$ so true for $n = 1$

Assume true for $n = k$:

$\frac{1}{3} \times \frac{3}{5} \times \frac{5}{7} \times \ldots \times \frac{2k-1}{2k+1} = \frac{1}{2k+1}$

Prove true for $n = k + 1$:

$\frac{1}{3} \times \frac{3}{5} \times \frac{5}{7} \times \ldots \times \frac{2k-1}{2k+1} \times \frac{2k+1}{2k+3}$

$= \frac{1}{2k+1} \times \frac{2k+1}{2k+3}$

$= \frac{1}{2k+3}$

$= \frac{1}{2(k+1)+1}$

Which is $\frac{1}{2n+1}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

4)

Prove true for $n = 1$:

LHS:

$$\sum_{r=0}^{1} \binom{1}{r} = \binom{1}{0} + \binom{1}{1} = 2$$

RHS: $2^1 = 2$ so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=0}^{k} \binom{k}{r} = 2^k$$

Prove true for $n = k + 1$:

$$\sum_{r=0}^{k+1} \binom{k+1}{r}$$

Since $\binom{k+1}{r} = \binom{k}{r} + \binom{k}{r-1}$

$$\sum_{r=0}^{k+1} \binom{k+1}{r} = \sum_{r=0}^{k+1} \binom{k}{r} + \sum_{r=0}^{k+1} \binom{k}{r-1}$$

Since $\binom{k}{r-1}$ is only defined for $r \geq 1$

$$\sum_{r=0}^{k+1} \binom{k}{r-1} = \sum_{r=1}^{k+1} \binom{k}{r-1} = \sum_{r=0}^{k} \binom{k}{r}$$

So

$$\sum_{r=0}^{k+1} \binom{k+1}{r} = \sum_{r=0}^{k+1} \binom{k}{r} + \sum_{r=0}^{k} \binom{k}{r}$$

$$= \sum_{r=0}^{k} \binom{k}{r} + \sum_{r=0}^{k} \binom{k}{r}$$

$$= 2^k + 2^k = 2 \times 2^k = 2^{k+1}$$

So true for $n = k + 1$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

5)

Prove true for $n = 1$:

LHS:

$$\sum_{r=1}^{1} \ln(2r-1) = \ln(2-1) = \ln 1 = 0$$

$\ln\left(\frac{(2n)!}{2^n n!}\right) = \ln\left(\frac{2!}{2^1 \times 1!}\right) = \ln\left(\frac{2}{2}\right) = \ln 1 = 0$ so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \ln(2r-1) = \ln\left(\frac{(2k)!}{2^n k!}\right)$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \ln(2r-1) = \sum_{r=1}^{k} \ln(2r-1) + \ln(2(k+1)-1)$$

$$= \ln\left(\frac{(2k)!}{2^k k!}\right) + \ln(2k+1)$$

$$= \ln\left(\frac{(2k+1)(2k)!}{2^k k!}\right)$$

$$= \ln\left(\frac{(2k+2)(2k+1)(2k)!}{(2k+2)2^k k!}\right)$$

$$= \ln\left(\frac{(2k+2)!}{2(k+1)2^k k!}\right)$$

$$= \ln\left(\frac{(2k+2)!}{2^{k+1}(k+1)!}\right)$$

$$= \ln\left(\frac{(2(k+1))!}{2^{k+1}(k+1)!}\right)$$

Which is $\ln\left(\frac{(2n)!}{2^n n!}\right)$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

6)

Prove true for $n = 1$:

LHS: $\mathrm{f}^1(x) = \frac{x}{x+1}$

RHS: $\frac{x}{1x+1} = \frac{x}{x+1}$ so true for $n = 1$

Assume true for $n = k$:

$\mathrm{f}^k(x) = \frac{x}{kx+1}$

Prove true for $n = k + 1$:

$\mathrm{f}^{k+1}(x) = \mathrm{f}\left(\mathrm{f}^k(x)\right) = \mathrm{f}\left(\frac{x}{kx+1}\right)$

$$= \frac{\frac{x}{kx+1}}{\frac{x}{kx+1}+1}$$

$$= \frac{x}{x+kx+1}$$

$$= \frac{x}{(k+1)x+1}$$

which $\frac{x}{kx+1}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

7)

Prove true for $n = 1$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}x}(x^1) = 1$

RHS: $n! = 1! = 1$ so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^k) = k!$

Prove true for $n = k + 1$:

$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(x^{k+1}) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^{k+1})\right)$

$= \frac{\mathrm{d}}{\mathrm{d}x}\left((k+1)x^k\right)$

$= (k+1)\frac{\mathrm{d}}{\mathrm{d}x}(x^k)$

$= (k+1)k!$

$= (k+1)!$ so true for $n = k + 1$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

8) a)

Prove true for $n = 1$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}\theta}(\sin a\theta) = a\cos a\theta$

RHS: $a\sin\left(a\theta + \frac{\pi}{2}\right)$

Since $\sin\left(x + \frac{\pi}{2}\right) = \cos x$

Let $x = a\theta$ so $a\sin\left(a\theta + \frac{\pi}{2}\right) = a\cos a\theta$ so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^k}{\mathrm{d}\theta^k}(\sin a\theta) = a^k\sin\left(a\theta + \frac{k\pi}{2}\right)$

Prove true for $n = k + 1$:

$\frac{\mathrm{d}^{k+1}}{\mathrm{d}\theta^{k+1}}(\sin a\theta) = \frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{\mathrm{d}^k}{\mathrm{d}\theta^k}(\sin a\theta)\right)$

$= \frac{\mathrm{d}}{\mathrm{d}\theta}\left(a^k\sin\left(a\theta + \frac{k\pi}{2}\right)\right)$

$= a^{k+1}\cos\left(a\theta + \frac{k\pi}{2}\right)$

$= a^{k+1}\sin\left(a\theta + \frac{k\pi}{2} + \frac{\pi}{2}\right)$

$= a^{k+1}\sin\left(a\theta + \frac{(k+1)\pi}{2}\right)$

Which is $a^n\sin\left(a\theta + \frac{n\pi}{2}\right)$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

8) b)

Prove true for $n = 1$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}\theta}(\ln(1+ax)) = \frac{a}{1+ax}$

RHS: $\frac{(-1)^{1+1}(1-1)!a^1}{(1+ax)^1} = \frac{a}{1+ax}$ so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(\ln(1+ax)) = \frac{(-1)^{k+1}(k-1)!a^k}{(1+ax)^k}$

Prove true for $n = k+1$:

$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(\ln(1+ax)) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^k}{\mathrm{d}x^k}(\ln(1+ax))\right)$

$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{(-1)^{k+1}(k-1)!a^k}{(1+ax)^k}\right)$

$= \frac{(-1)^{k+1}(k-1)!a^k \times -ka}{(1+ax)^k}$

$= \frac{(-1)(-1)^{k+1} \times k(k-1)! \times a \times a^k}{(1+ax)^k}$

$= \frac{(-1)^{k+2}k!a^{k+1}}{(1+ax)^{k+1}}$

$= \frac{(-1)^{(k+1)+1}(k+1-1)!a^{k+1}}{(1+ax)^{k+1}}$

Which is $\frac{(-1)^{n+1}(n-1)!a^n}{(1+ax)^n}$ with $n = k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$

9) a)

Prove true for $n = 1$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{ax+b}) = a\mathrm{e}^{ax+b}$

RHS: $a^1\mathrm{e}^{ax+b}$ so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(\mathrm{e}^{ax+b}) = a^k\mathrm{e}^{ax+b}$

Prove true for $n = k + 1$:

$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(\mathrm{e}^{ax+b}) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^k}{\mathrm{d}x^k}(\mathrm{e}^{ax+b})\right)$

$= \frac{\mathrm{d}}{\mathrm{d}x}(a^k\mathrm{e}^{ax+b})$

$= a^{k+1}\mathrm{e}^{ax+b}$

Which is $a^n\mathrm{e}^{ax+b}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

9) b)

Let $\mathrm{f}(x) = \mathrm{e}^{ax+b}$, so $\mathrm{f}(0) = \mathrm{e}^b$

Let $\mathrm{f}'(x) = a\mathrm{e}^{ax+b}$, so $\mathrm{f}'(0) = a\mathrm{e}^b$

Let $\mathrm{f}''(x) = a^2\mathrm{e}^{ax+b}$, so $\mathrm{f}''(0) = a^2\mathrm{e}^b$

$\therefore \mathrm{e}^{ax+b} = \mathrm{e}^b + a\mathrm{e}^b x + \frac{a^2\mathrm{e}^b}{2}x^2$

10)

Prove true for $n = 1$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^x \cos\left(\sqrt{3}x\right)\right) = \mathrm{e}^x \cos\left(\sqrt{3}x\right) - \sqrt{3}\mathrm{e}^x \sin\left(\sqrt{3}x\right)$

$= \mathrm{e}^x\left[\cos\left(\sqrt{3}x\right) - \sqrt{3}\sin\left(\sqrt{3}x\right)\right]$

$= 2\mathrm{e}^x\left[\frac{1}{2}\cos\left(\sqrt{3}x\right) - \frac{\sqrt{3}}{2}\sin\left(\sqrt{3}x\right)\right]$

$= 2\mathrm{e}^x\left[\cos\left(\frac{\pi}{3}\right)\cos\left(\sqrt{3}x\right) - \sin\left(\frac{\pi}{3}\right)\sin\left(\sqrt{3}x\right)\right]$

$= 2\mathrm{e}^x \cos\left(\frac{\pi}{3} + \sqrt{3}x\right)$

$= 2\mathrm{e}^x \cos\left(\sqrt{3}x + \frac{\pi}{3}\right)$

RHS: $2^1\mathrm{e}^x \cos\left(\sqrt{3x} + \frac{\pi}{3} \times 1\right)$ so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^k}{\mathrm{d}x^k}\left(\mathrm{e}^x \cos\left(\sqrt{3}x\right)\right) = 2^k\mathrm{e}^x \cos\left(\sqrt{3}x + \frac{\pi}{3}k\right)$

Prove true for $n = k + 1$:

$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}\left(\mathrm{e}^x \cos\left(\sqrt{3}x\right)\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^k}{\mathrm{d}x^k}\left(\mathrm{e}^x \cos\left(\sqrt{3}x\right)\right)\right)$

$= \frac{\mathrm{d}}{\mathrm{d}x}\left(2^k\mathrm{e}^x \cos\left(\sqrt{3}x + \frac{\pi}{3}k\right)\right)$

$= -2^k\mathrm{e}^x \times \sqrt{3}\sin\left(\sqrt{3}x + \frac{\pi}{3}k\right) + 2^k\mathrm{e}^x \cos\left(\sqrt{3}x + \frac{\pi}{3}k\right)$

$= 2^k\mathrm{e}^x\left[\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right) - \sqrt{3}\sin\left(\sqrt{3}x + \frac{\pi}{3}k\right)\right]$

$= 2 \times 2^k\mathrm{e}^x\left[\frac{1}{2}\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right) - \frac{\sqrt{3}}{2}\sin\left(\sqrt{3}x + \frac{\pi}{3}k\right)\right]$

$= 2^{k+1}\mathrm{e}^x\left[\cos\left(\frac{\pi}{3}\right)\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right) - \sin\left(\frac{\pi}{3}\right)\sin\left(\sqrt{3}x + \frac{\pi}{3}k\right)\right]$

$= 2^{k+1}\mathrm{e}^x\left[\cos\left(\frac{\pi}{3} + \left(\sqrt{3}x + \frac{\pi}{3}k\right)\right)\right]$

$= 2^{k+1}\mathrm{e}^x\left[\cos\left(\sqrt{3}x + \frac{\pi}{3}(k+1)\right)\right]$

Which is $2^n\mathrm{e}^x \cos\left(\sqrt{3}x + \frac{\pi}{3}n\right)$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

11)

Prove true for $n = 2$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}(\sinh x)\right) = \frac{\mathrm{d}}{\mathrm{d}x}(\cosh x) = \sinh x$ which is the RHS so true for $n = 2$

Assume true for some even integer $n = k$ where $k \geq 2$:

$\frac{\mathrm{d}^k y}{\mathrm{d}x^k} = \frac{\mathrm{d}^k}{\mathrm{d}x^k}(\sinh x) = \sinh x$

Prove true for $n = k + 2$:

$\frac{\mathrm{d}^{k+2}}{\mathrm{d}x^{k+2}}(\sinh x) = \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\mathrm{d}^k}{\mathrm{d}x^k}(\sinh x)\right)$

$= \frac{\mathrm{d}^2}{\mathrm{d}x^2}(\sinh x)$

$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}(\sinh x)\right)$

$= \frac{\mathrm{d}}{\mathrm{d}x}(\cosh x)$

$= \sinh x$

As the statement is true for $n = 2$, and if the statement is true for $n = k$ then the statement is true for $n = k + 2$, so the statement is true for all even integers $n \geq 2$

12)

Prove true for $n = 1$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}x}(x \sinh x) = x \cosh x + \sinh x$

RHS: $x \cosh x + (2 - 1) \sinh x = x \cosh x + \sinh x$ so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^{2k-1}y}{\mathrm{d}x^{2k-1}} = x \cosh x + (2k - 1) \sinh x$

Prove true for $n = k + 1$:

$\frac{\mathrm{d}^{2(k+1)-1}y}{\mathrm{d}x^{2(k+1)-1}} = \frac{\mathrm{d}^{2k+1}y}{\mathrm{d}x^{2k+1}}$

$= \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\mathrm{d}^{2k-1}y}{\mathrm{d}x^{2k-1}}\right)$

$= \frac{\mathrm{d}^2}{\mathrm{d}x^2}(x \cosh x + (2k - 1) \sinh x)$

$= \frac{\mathrm{d}}{\mathrm{d}x}(x \sinh x + \cosh x + (2k - 1) \cosh x)$

$= \frac{\mathrm{d}}{\mathrm{d}x}(x \sinh x + 2k \cosh x)$

$= x \cosh x + \sinh x + 2k \sinh x$

$= x \cosh x + (2k + 1) \sinh x$

$= x \cosh x + (2(k + 1) - 1) \sinh x$

Which is $x \cosh x + (2n - 1) \sinh x$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

13)

Prove true for $n = 1$:

LHS: $\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^x \cosh x) = \mathrm{e}^x \sinh x + \mathrm{e}^x \cosh x = \mathrm{e}^x(\cosh x + \sinh x)$

RHS: $2^0\mathrm{e}^x(\cosh x + \sinh x) = \mathrm{e}^x(\cosh x + \sinh x)$ so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^k y}{\mathrm{d}x^k} = 2^{k-1}\mathrm{e}^x(\cosh x + \sinh x)$

Prove true for $n = k + 1$:

$\frac{\mathrm{d}^{k+1} y}{\mathrm{d}x^{k+1}} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^k y}{\mathrm{d}x^k}\right)$

$= \frac{\mathrm{d}}{\mathrm{d}x}\left(2^{k-1}\mathrm{e}^x(\cosh x + \sinh x)\right)$

$= 2^{k-1}\mathrm{e}^x(\sinh x + \cosh x) + 2^{k-1}\mathrm{e}^x(\cosh x + \sinh x)$

$= 2 \times 2^{k-1}\mathrm{e}^x(\cosh x + \sinh x)$

$= 2^k\mathrm{e}^x(\cosh x + \sinh x)$

$= 2^{(k+1)-1}\mathrm{e}^x(\cosh x + \sinh x)$

Which is $2^{n-1}\mathrm{e}^x(\cosh x + \sinh x)$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

14)

Prove true for $n = 1$:

LHS: x

RHS: $\frac{x(1-(-x)^1)}{1+x} = \frac{x(1+x)}{1+x} = x$ so true for $n = 1$

Assume true for $n = k$:

$f_k(x) = \frac{x(1-(-x)^k)}{1+x}$

Prove true for $n = k + 1$:

$f_{k+1}(x) = f_k(x) + (-1)^{k+1-1}x^{k+1}$

$= \frac{x(1-(-x)^k)}{1+x} + (-1)^k x^{k+1}$

$= \frac{x(1-(-x)^k)}{1+x} + (-x)^k x$

$= x\left(\frac{1-(-x)^k}{1+x} + (-x)^k\right)$

$= x\left(\frac{1-(-x)^k}{1+x} + \frac{(1+x)(-x)^k}{1+x}\right)$

$= x\left(\frac{1-(-x)^k+(1+x)(-x)^k}{1+x}\right)$

$= x\left(\frac{1-(-x)^k+(-x)^k+x(-x)^k}{1+x}\right)$

$= x\left(\frac{1+x(-x)^k}{1+x}\right)$

$= x\left(\frac{1-(-x)^{k+1}}{1+x}\right)$

$= \frac{x(1-(-x)^{k+1})}{1+x}$

Which is $\frac{x(1-(-x)^n)}{1+x}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

15)

Prove true for $n = 1$:

LHS: $(1 + x)^1 = 1 + x$

RHS: $1 + {}^1C_1x = 1 + x$ so true for $n = 1$

Assume true for $n = k$:

$(1 + x)^k \equiv 1 + {}^kC_1x + {}^kC_2x^2 + \cdots + {}^kC_kx^k$

Prove true for $n = k + 1$:

$(1 + x)^{k+1} = (1 + x)(1 + x)^k$

$= (1 + x)\left(1 + {}^kC_1x + {}^kC_2x^2 + \cdots + {}^kC_kx^k\right)$

$= 1 + {}^kC_1x + {}^kC_2x^2 + \cdots + {}^kC_kx^k + x + {}^kC_1x^2 + {}^kC_2x^3 + \cdots + {}^kC_kx^{k+1}$

$= 1 + \left(1 + {}^kC_1\right)x + \left({}^kC_1 + {}^kC_2\right)x^2 + \left({}^kC_2 + {}^kC_3\right)x^3 + \cdots + \left({}^kC_{k-1} + {}^kC_k\right)x^k + \left({}^kC_k\right)x^{k+1}$

But Pascal's Rule gives: ${}^kC_{r-1} + {}^kC_r = {}^{k+1}C_r$

So that

$(1 + x)^{k+1}$

$= 1 + {}^{k+1}C_1x + {}^{k+1}C_2x^2 + {}^{k+1}C_3x^3 + \cdots + {}^{k+1}C_kx^k + \left({}^{k+1}C_{k+1}\right)x^{k+1}$

Which is $(1 + x)^n \equiv 1 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$